

ARYAN SOMAYAJULA

AI Engineer · LLM Systems · Agentic Pipelines · Edge AI · Android

London, UK | somayajulaaryan@gmail.com | +44 7407 750520 | LinkedIn | GitHub
UK Post-Study Work Visa (Jan 2027) • Open to Skilled Worker Sponsorship

Summary

AI Engineer with MSc Data Science & Analytics (Merit, Royal Holloway) specialising in LLM systems, agentic pipelines, RAG architectures, and edge AI inference. Built production-grade, privacy-preserving multi-agent systems running fully locally on embedded hardware. Background spans Android development, ML engineering, and technical team leadership. Immediately available.

Technical Skills

LLM & Agents	LLMs (Phi-3 Mini, LLaMA 3 8B, Whisper), Agentic Workflows, RAG, Context Engineering, Prompt Engineering, LLM Evals, 4-bit Quantisation, Semantic Extraction
ML & Data	Scikit-learn, LightGBM, Reinforcement Learning, GMMs, NILM, Time-Series Forecasting, Signal Processing, Pandas, NumPy, SQL, InfluxDB, SQLite
Backend & Infra	Python, FastAPI, Mosquitto MQTT, REST APIs, Docker Compose, AWS S3, Edge Computing
Android & Mobile	Java, Kotlin, Android SDK, REST API Integration, AWS S3, UI/UX, Razorpay
Embedded / OS	Buildroot, Embedded Linux, ARM64, Raspberry Pi, Wayland/Weston, systemd

Projects & Research

NestShift OS — Local-First Agentic Smart Home AI · Independent Research 2025 – Present

- Architected a **three-agent LLM system** (Energy Brain / Automation Brain / System Brain) running entirely on Raspberry Pi — Phi-3 Mini + LLaMA 3 8B, 4-bit quantised, **zero cloud dependency**.
- Built semantic state-retrieval pipeline (SQLite + InfluxDB + RAG) injecting live sensor data into context; FastAPI + Mosquitto MQTT event-driven backend; sub-200ms intent parsing via Whisper STT.
- Designed **safe constraint system** — immutable rule layer overriding unsafe agent actions; integrated Octopus Agile UK tariff API for real-time energy optimisation and NILM load disaggregation.
- Delivered full technical whitepaper, investor pitch deck, and academic research companion (4 publishable gaps identified: NILM accuracy, EU AI Act compliance, federated learning, edge model drift).

COVID-19 Detection from Cough Sounds · MSc Dissertation — Royal Holloway 2024

- ML pipeline combining MFCC/spectral feature extraction, Reinforcement Learning, and Random Forest classification on crowdsourced audio — **96% classification accuracy**.

Experience

Team Lead — Operations · Benugo | Waterloo Station, London Oct 2024 – Present

- Lead a **15+ person team** at London's highest-footfall transport hub; manage incident response, real-time resource allocation, performance reviews, and structured coaching.

Android App Developer Intern · Grupo Technologies Pvt Ltd | Bangalore, India Aug 2022 – Aug 2023

- Built **2 production Android apps** in Java; integrated REST APIs, AWS S3 media storage, and Razorpay payment gateway; owned UI/UX design end-to-end.

Frontend Developer Intern · Webingo Infotech Pvt Ltd | Kolkata, India Apr – Jun 2022

- Delivered UI/UX across 3 Android apps; awarded Certificate of Achievement for team contribution.

Education

MSc Data Science & Analytics — Merit (2:1) · Royal Holloway, University of London 2023 – 2024

Bachelor of Computer Applications · Birla Institute of Technology, Mesra 2020 – 2023